

Music Sound Interpreter

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Team #6

One Man Army

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The Rundown

- Who is the client?
- What is the project?
- Why is this being done?
- How will this project be done?
- How much time will it take?
- How much is this project going to cost?

The Client

- My client is Rosemary
 - Has Down Syndrome
 - Is deaf
 - Communicates by sign language
 - Spends some of her time at Bridgewell in Lynn

Some Facts about Down Syndrome

- Down Syndrome is a genetic disorder involving the 21st chromosome
- About 1 in 700 people are born with DS
- 400,000 diagnosed people in the U.S.
- Most prominent type is Trisomy 21
 - Accounts for 95% of all cases
 - Cells have an extra copy of chromosome 21

Effects of Down Syndrome

- Effects vary from person to person
- Possible for diagnosed people to lead normal lives
- Common effects include:
 - Physical disfigurement
 - Slow mental development
 - Increased risk of health issues

The Problem

- Bridgewell hosts learning activities such as concerts
- Since Rosemary is deaf, she feels the speakers to understand what's going on
- In doing so, she is physically left out of her groups

The Solution: My Project

- Lap pad that senses audio signals then activates visual effects on one side and vibrational effects on the other
- Contains recorded responses such as: "That's awesome!"
- Using the Music Sound Interpreter,
she will able to join her groups extensively

Functions

- Visual and vibrational effects controlled by frequency and strength of audio signals
- Visual components consist of a rectangular LED grid
 - Each column represents a different frequency range
 - As intensity changes, LEDs fade in and out
- Vibrational effects activated similarly

Bonus Purposes

- Can be used to teach individuals how to control their voices
 - Instructor would say something to be copied,
then the user would then try to match the instructor
- Could also be used as decorative piece

Designing for Everyone

- Rosemary will most likely not be the only user
- Should be designed so it's usable by anyone
- Design factors include:
 - Safety
 - Accessibility
 - Simplicity

Physical Specifications

- About as big as an open notebook (12"x21"x1")
- Weight = 5 lbs
 - The lighter, the better
- Cushioned
- Shatterproof
- Waterproof
- Easily cleanable
- Cased in flexible material

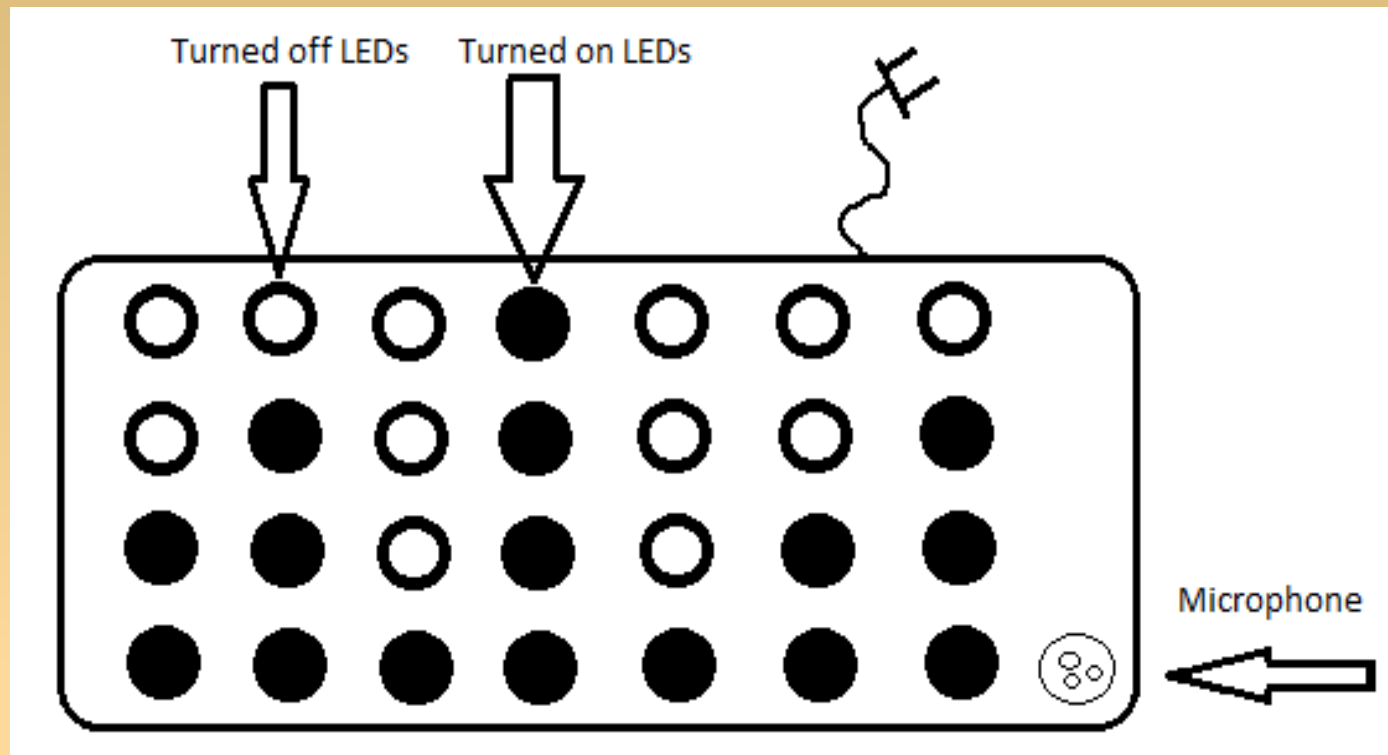
Electrical Specifications

- Powered via wall outlet
- Includes a function toggle switch
 - Only vibrations, only lights, both on, both off
- Automatic shut-off feature
- Low voltage, about 12V maximum
- Medium current, about 2A maximum

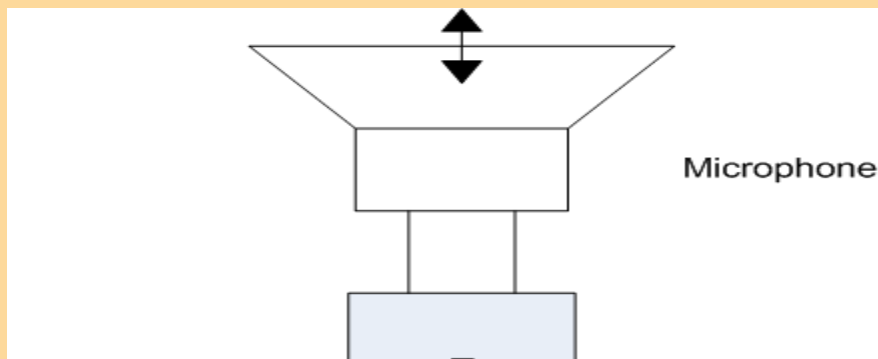
Effect Specs

- Uses calming colors such as pink, purple and green
 - Avoid using red, yellow, and orange
- Uses smooth visual components
 - No fast-blinking lights
- Intensity of lights and vibrations shouldn't be strong

Functional Diagram



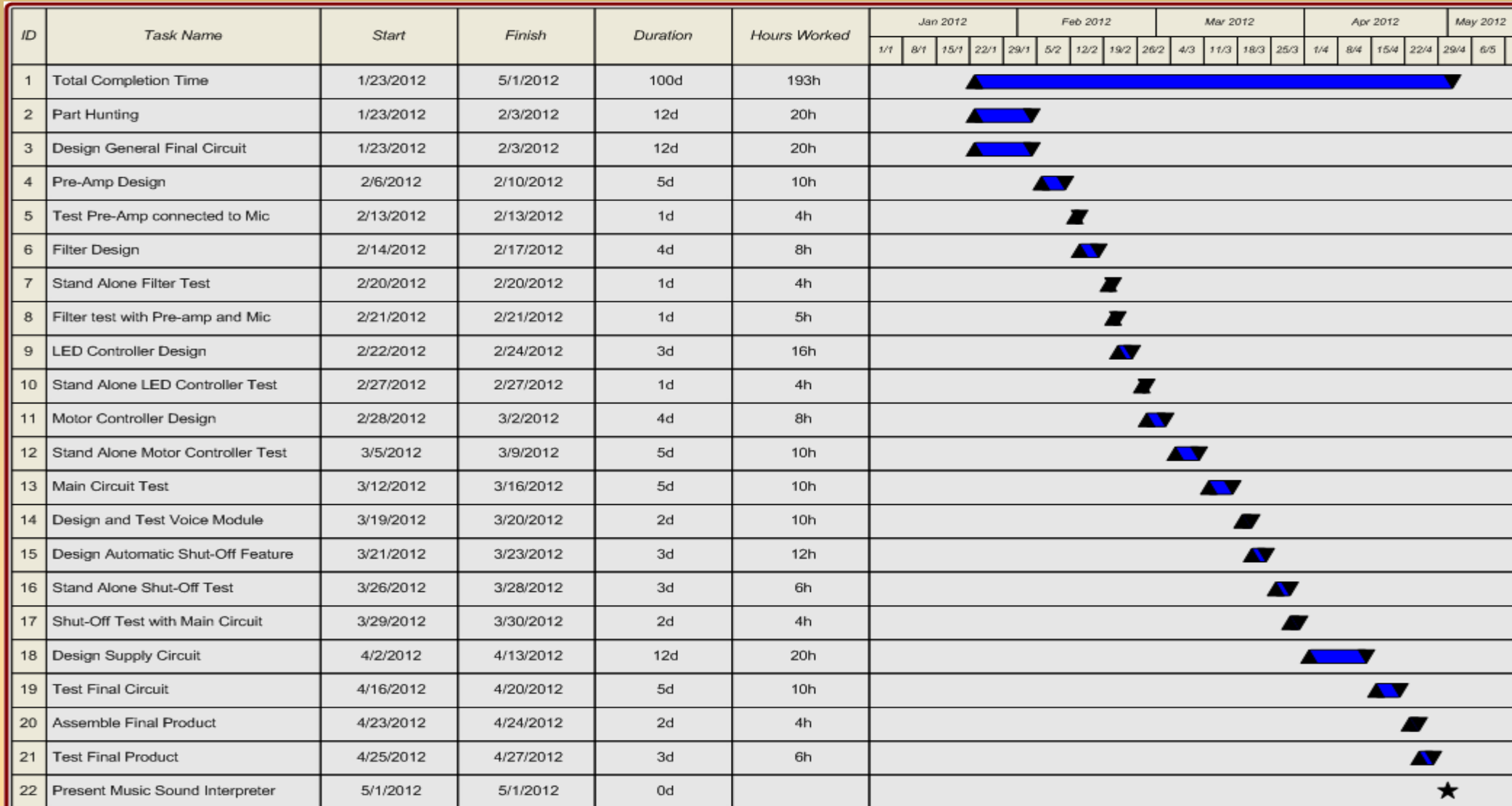
Block Diagram



Creation Procedure

- Design parts from beginning of product activation towards output
- Test parts in descending order
 - First tests are stand alone
 - Second are with previous parts included

Gantt Chart



Time Analysis

- Beginning Date: January 23, 2012
- Ending Date: May 1, 2012
- Duration: 100 days
- Time spent on project: 193 hours
- Designing each block mostly takes up a week
- Testing usually takes a few days
- Work between 2-4 hours per day

Market Research

- Most products are very expensive
- Not many found that have similar functions
- Most products have simple, “discrete” functions
- Very few are sound controlled
- Potential for large market, even outside of assistive technology

Main Competing Product

Interactive Fanlight

- Displays lights in a fan-like manner
- Automatic and sound controlled modes
- Multiple display styles within each mode
 - Only amplitude dependent
 - Priced at \$2,675.00

Labor and Overhead

- Average time worked per week: 13.5 hours
- 14 week long project
- Labor Cost = \$5,724
- Overhead Cost= \$2,919

Parts and Prices

Total Cost

- Direct Labor: \$5,724
- Overhead: \$2,919
- Parts: \$325
- **Total Cost of Project: \$8,962**

Final Words

Any Questions?

Thanks everyone for your time!